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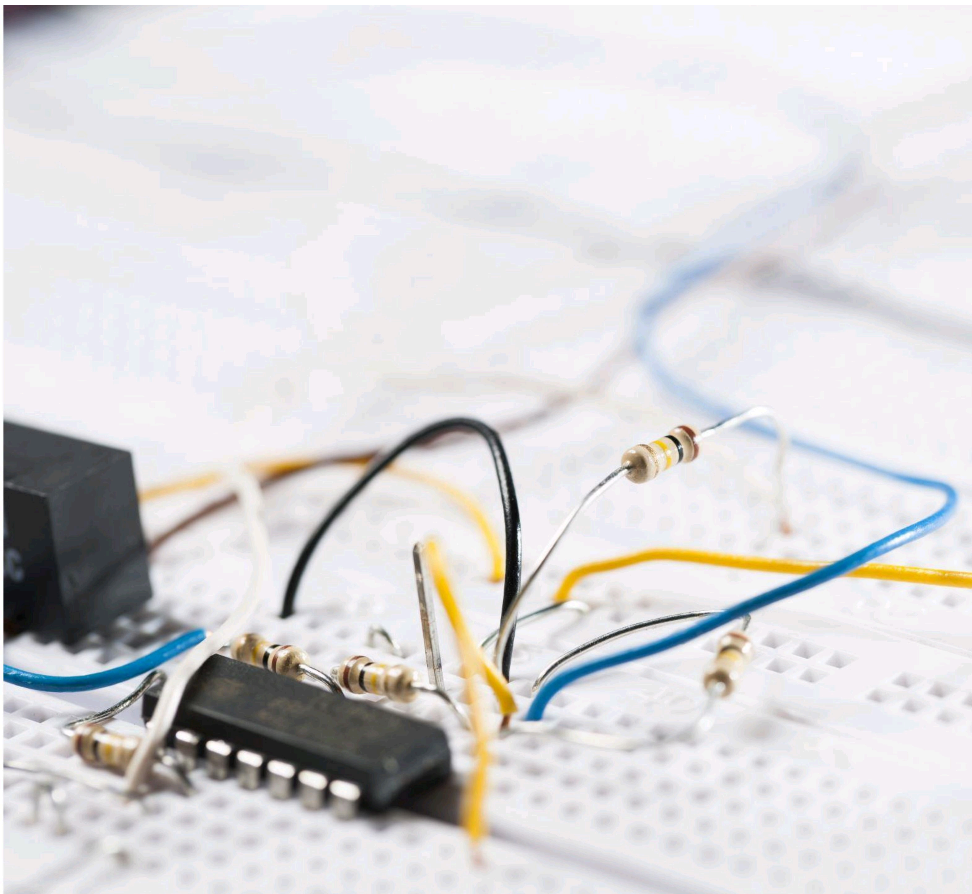


Introduction

- The Silicon Controlled Rectifier (SCR) is a widely used semiconductor device in power electronics for controlling the flow of electrical power. It acts as a switch that can be turned on by applying a suitable gate pulse and remains conducting until the current falls below its holding value. To achieve accurate and reliable control over the firing angle of the SCR, a proper triggering circuit is essential.

- A UJT (Unijunction Transistor) relaxation oscillator is commonly employed for generating triggering pulses for SCRs due to its simplicity, stability, and ability to produce sharp and consistent pulses. In a synchronized UJT relaxation oscillato rcircuit, the oscillator is synchronized with the AC supply voltage, ensuring that the firing pulses are applied at a controlled point in each half cycle. This synchronization allows precise phase control of the SCR, enabling smooth variation of power delivered to the load.

LITERATURE SURVEY

- **Md. Moyeed Abrar (2018)** conducted an experimental investigation of synchronized UJT trigger circuits using the unijunction transistor (UJT 2N2646) to generate precise gate pulses for SCR triggering. His work, published in the International Journal of Advanced Technology and Engineering Exploration, demonstrated how synchronization of the UJT relaxation oscillator with the AC supply ensures stable phase control and reliable SCR turn-on characteristics.
- **Muhammad H. Rashid (2017),** in his book Power Electronics: Circuits, Devices and Applications (Pearson Education), described the operation of UJT relaxation oscillators used as triggering sources for SCRs. The book explains how the charging and discharging of the timing capacitor through the UJT emitter lead to a sawtooth waveform that can be synchronized with the AC signal for controlled rectification and phase-angle control applications.
- **M. D. Singh and K. B. Khanchandani (2011),** in their book Power Electronics (Tata McGraw-Hill Publications), discussed the use of synchronized UJT relaxation oscillators for SCR gate triggering in controlled rectifiers. The authors emphasized circuit design aspects, frequency stability, and synchronization with the AC mains for improved firing angle accuracy.

ProblemStatement

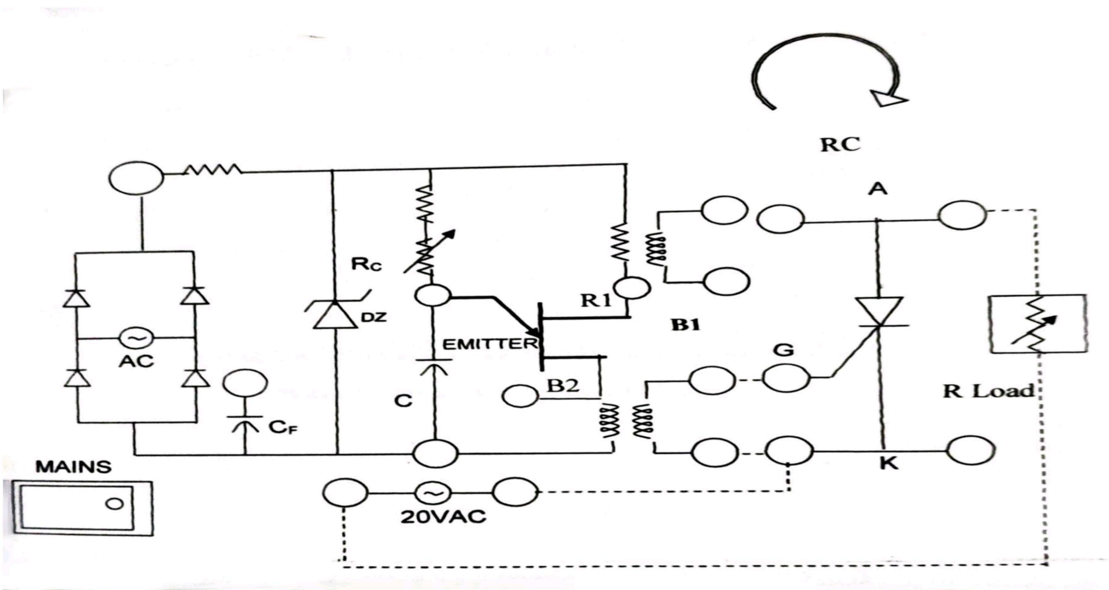
In conventional triggering methods, variations in supply voltage and component tolerances can cause unstable firing angles and poor synchronization with the AC mains. To overcome this, a synchronized UJT relaxation oscillator is employed, which generates sharp and consistent gate pulses at controlled intervals, synchronized with the AC waveform.

This circuit aims to achieve accurate control over the firing angle of the SCR, ensuring smooth regulation of power delivered to the load. By adjusting the oscillator's timing components, the firing delay can be varied across the AC cycle, allowing efficient control of output voltage and current.

Objective

1. Precise SCR triggering synchronized with the AC waveform .
2. Controlled turn on timing of SCR for Power regulation.

Circuit diagram



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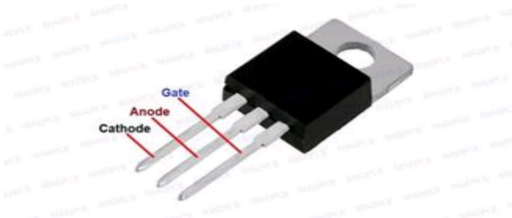


1.SCR(Silicon controlled rectifier):
purpose: Acts as the main power control device, conducts when triggered.
Specifications:400V,IT=4A

2.UJT (Unijunction Transistor):
Purpose: Acts as the relaxation oscillator to generate trigger pulses.
Specifications:VB2B1=10-20V

3.CAPACITOR:
Purpose: charges and discharges to control UJT triggering time.
Specifications:0.1Micro farad-0.47micro farad,100V

4.RESISTOR
Purpose: Adjust the capacitor charging rate: controls firing angle.
Specifications:100kilo ohm potentiometer.

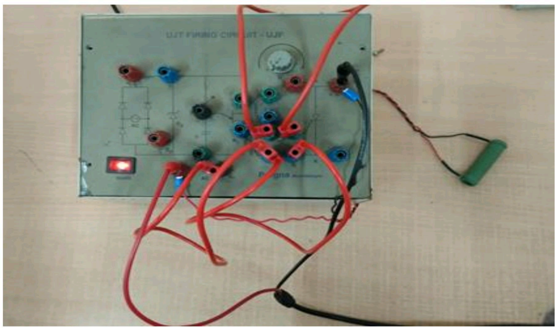
5.zener diode
Purpose: Regulates the Dc supply voltage to the UJT circuit; maintains a stable reference voltage despite supply variations.
Specifications:12V,0.5W

WORKING:

When AC supply is applied, a part of it is rectified and regulated by the Zener diode (Dz) to supply the UJT relaxation oscillator. The capacitor (C) charges through the variable resistor (R_1). When the capacitor voltage reaches the peak-point voltage of the UJT, the UJT switches ON and the capacitor discharges rapidly through the primary winding of the pulse transformer. This induces a sharp triggering pulse in the secondary winding, which is applied to the gate (G) of the SCR, turning it ON.

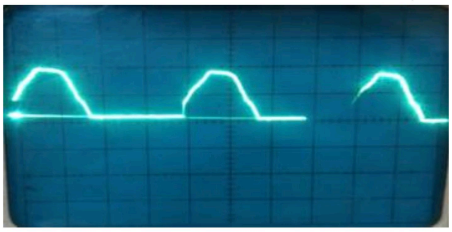
By adjusting the resistor R_1, the charging time of the capacitor changes, thereby varying the firing angle (α) of the SCR. The process repeats in every half cycle, producing synchronized gate pulses with the AC supply. This allows smooth control of power to the load such as lamps, heaters, or motors.

PICTURE OF MODEL



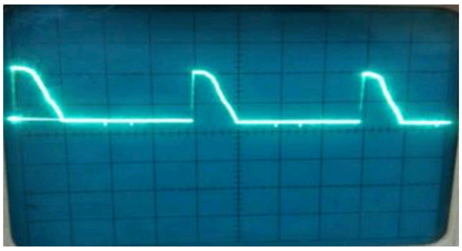
Output

Waveform of Half wave Output Angle=0°



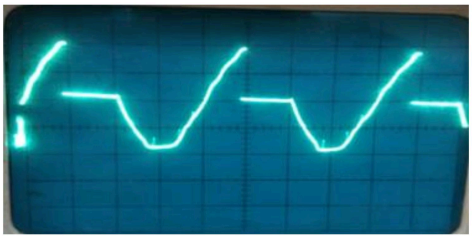
Vout

Waveform of load
Angle=90°



Vload

Waveform across load



Vscr

APPLICATIONS

- Speed control of AC and DC motors
- Light dimming and heating control
- Controlled rectifiers
- Automatic power switching
- Pulse generators
- Sawtooth wave generator
- Switch-mode power supplies (SMPS)

ADVANTAGES

- Precise and stable triggering
- Accurate phase control
- Efficient and robust triggering pulse
- Sharp, fast-rising pulses
- Minimizes power dissipation
- Circuit simplicity and cost-effectiveness
- Overcomes limitations of simpler firing circuits

conclusion

The SCR turn-on circuit using a synchronized UJT relaxation oscillator is an efficient and reliable method for controlling the firing angle of an SCR in AC circuits. In this circuit, the UJT relaxation oscillator generates sharp and consistent triggering pulses that are synchronized with the AC supply voltage. By adjusting the resistor and capacitor values in the timing network, the firing angle of the SCR can be varied, allowing precise control of the output power delivered to the load. This results in smooth and stable operation with accurate phase control. Due to its simplicity, reliability, and effectiveness, this method is widely used in applications such as light dimmers, controlled rectifiers, and motor speed control systems.

References

1.M.D. Moyeed Abrar published in IJATEE in the year 2018 that states developed and tested synchronized UJT trigger circuit for SCR turn on improved phase controlled stability.

2.M.H. Rashid has published a book name power electronics in the year 2017 that explains UJT relaxation oscillator working and SCR triggering method for phase angle method.

3.MD Singh and K.B. khanchandani in 2011 has published a book named power electronics which elaborates discussion on the synchronized UJT oscillator firing circuit for SCR and phase controlled in AC rectifier.